27. Given the generator matrix $G = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{pmatrix}$ .corresponding to the encoding function $e_G : B^4 \to B^7$ .find the corresponding parity check matrix and use it to decode the following received words and hence to find the original message 110 0001 , 1110111 , 0010001 , 0011100. 15 3 4

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Reg. No															

B.Tech/M.Tech (Integrated) DEGREE EXAMINATION, JULY 2025

Fifth Semester

21MAB302T - DISCRETE MATHEMATICS

(For the candidates admitted during the academic year 2021-2022 to 2024-2025)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.
ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

Marks BL CO

1. $A-(B\cap C)=$ 1 1 1
(A) $(A-B)\cup(A-C)$ (B) $(A-B)\cap(A-C)$
(C) $(A-B)\cup(B-C)$ (D) $(A-B)\cap(B-C)$

2. If A is a non empty set with n elements, then the number of possible relations on the set A is 1 2 1
(A) 2^n (B) 2^{n-1}
(C) 2^{n^2} (D) 2^{n+1}

3. Let R={(1,1),(2,2),(3,3)} be a relation on A={1,2,3}. Then R is 1 2 1
(A) Reflexive only (B) Symmetric only
(C) Transitive only (D) An equivalence relation

4. If f and g are onto then the composition $g\circ f$ is 1 1 1
(A) One-one (B) Many one
(C) Onto (D) Into

5. In how many ways the letters of word "STRESS" can be arranged? 1 2 2
(A) 360 (B) 720
(C) 240 (D) 120

6. If $gcd(a,b)=1$ then $gcd(ka,kb)=$ 1 2 2
(A) 1 (B) k
(C) k^2 (D) $2k$

7. Every integer $n>1$ can be written uniquely as a product of __________ 1 1 2
(A) any integers (B) prime numbers
(C) composite numbers (D) only odd integers

8. If 36 pigeons are accommodated in 4 pigeonholes, the least number of pigeons accommodated in one pigeon hole is 1 2 2
(A) 8 (B) 9
(C) 10 (D) 11

9. A premise may be introduced at any steps in the derivation is known as 1 1 3
(A) Rule T (B) Rule P
(C) Rule C (D) Rule CP

10. $p\rightarrow q$ is logically equivalent to 1 1 3
(A) $\neg p\vee q$ (B) $p\vee\neg q$
(C) $\neg p\vee\neg q$ (D) $p\vee q$

11. Dual of $(p \wedge q) \vee r$ 1 2 3

(A) $(p \vee q) \vee r$ (B) $(p \wedge q) \wedge r$
(C) $(p \wedge q) \vee r$ (D) $(p \vee q) \wedge r$

12. $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$ is a 1 2 3

(A) Tautology (B) Contradiction
(C) Contingency (D) Negation

13. If a, b are the elements of a group G then $(a * b)^{-1}$ 1 1 4

(A) $a^{-1} * b^{-1}$ (B) $b^{-1} * a^{-1}$
(C) $b * a$ (D) $a * b$

14. If α and β are the elements of the symmetric group S_4 and are given by $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 2 & 1 \end{pmatrix}$, $\beta = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 3 & 1 \end{pmatrix}$, then $\alpha\beta$ is given by 1 2 4

(A) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}$ (B) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 1 & 4 & 2 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}$ (D) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$

15. The number of identity elements in a group is 1 1 4

(A) 0 (B) 1
(C) 2 (D) 3

16. The hamming weight of $x = 111010$ 1 2 4

(A) 2 (B) 3
(C) 4 (D) 6

17. A tree with 10 vertices has _______ edges 1 2 5

(A) 8 (B) 9
(C) 10 (D) 11

18. A path in a connected graph $G = (V, E)$ is called Hamiltonian path if 1 1 5

(A) It includes every edge exactly once (B) It includes every vertex exactly once
(C) It includes every edge exactly twice (D) It includes every vertex exactly twice

19. The chromatic number of a complete bipartite graph $K_{4,3}$ 1 2 5

(A) 2 (B) 3
(C) 5 (D) 6

20. If every vertex of a simple graph has the same degree then the graph is called 1 1 5

(A) A complete graph (B) A regular graph
(C) Path (D) Tree

PART - B (5 × 8 = 40 Marks) Marks BL CO

Answer **all** Questions

21. (a) Let $A = \{x \in R | x \neq 2\}$, $B = \{x \in R | x \neq 1\}$. Define $f : A \rightarrow B$ by $f(x) = \frac{x}{x-2}$. Prove f is 1-1 and onto. Also find f^{-1}. 8 3 1

(OR)

(b) Find the minsets and maxsets generated by $A = \{1, 7, 8\}$, $B = \{1, 6, 9, 10\}$, $C = \{1, 9, 10\}$ where $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and prove that minsets form partition of U.

22. (a) Use Euclidean algorithm to find $gcd(12345, 54321)$ and express linear combination of the given numbers. 8 3 2

(OR)

(b) Find the number of integers between 1 and 250 (both inclusive) that are not divisible by any of the integers $2, 3, 5$ and 7

23. (a) Use mathematical induction to prove that $(3^n + 7^n - 2)$ is divisible by 8 for $n \geq 1$ 8 3 3

(OR)

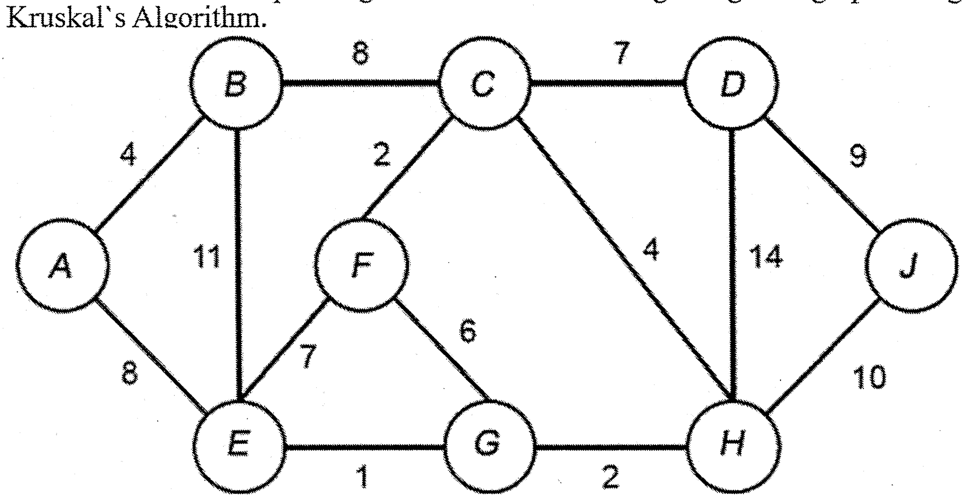
(b) Show that the following set of premises is inconsistent
If Rama gets his degree, he will go for a job
If he goes for a job, he will get married soon
If he goes for higher study, he will not get married
Rama gets his degree and goes for higher study.

24. (a) Prove that every subgroup of a cyclic group is cyclic. 8 3 4

(OR)

(b) Find the code words generated by the encoding function $e : B^3 \rightarrow B^6$ with respect to parity check matrix $H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

25. (a) Find the minimum spanning tree for the following weighted graph using Kruskal's Algorithm. 8 3 5



(OR)

(b) Prove that maximum number of edges in a simple disconnected graph G with n vertices and k components is $\frac{(n-k)(n-k+1)}{2}$

PART - C (1 × 15 = 15 Marks) Marks BL CO

Answer **any 1** Questions

26. Find the transitive closure of $R = \{(1,2), (2,1), (2,3), (3,4), (4,2), (4,4), (5,1), (5,5)\}$ on $A = \{1, 2, 3, 4, 5\}$ using Warshall's Algorithm 15 4 1